

0.1 Gaussian Response Models

0.1.1 Intro

EXAMPLE 0.1.1: STAT 230 and 231 Final Grades

No.	S230	S231
1	76	76
2	77	79
3	57	54
4	75	64
5	74	64
6	60	60
7	81	85
8	86	82
9	96	88
10	79	72

No.	S230	S231
11	87	76
12	71	50
13	63	75
14	77	72
15	96	84
16	65	69
17	71	43
18	66	60
19	90	96
20	50	50

No.	S230	S231
21	98	83
22	80	88
23	67	52
24	78	75
25	100	99
26	94	94
27	83	83
28	51	37
29	77	90
30	77	67

- Why might we be interested in collecting data such as these?
- What might be a reasonable choice for the target and study population?
- What are the variates? What type are they?
- What is the explanatory variate? What is the response variate?
- How do we summarize these data numerically and graphically?
- What model could we use to analyse these data?

0.1.2 Sample Correlation

Recall that the sample correlation is a numerical measure of the linear relationship between two variates. It is defined as

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$$

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n(\bar{x})^2$$

$$S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}$$

$$S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n y_i^2 - n(\bar{y})^2$$

Recall that $-1 \leq r \leq 1$

EXAMPLE 0.1.2: Sample Correlation for STAT 230/231 Final Grades

Let x be the STAT 230 final grade, and y be the STAT 231 final grade. For these data, we have

$$S_{xx} = 5135.8667 \quad S_{xy} = 5106.8667 \quad S_{yy} = 7585.3667$$

Thus,

$$r = \frac{5106.8667}{\sqrt{(5135.8667)(7585.3667)}} = 0.82$$

Since r is close to 1, we would say that there is a strong positive linear relationship between STAT 230 and STAT 231 final grades.

0.1.3 Least Squares Estimates

Fitting a Straight Line: Least Squares Approach

To determine the fitted line $y = \alpha + \beta x$, which minimizes the sum of the squares of the distances between the observed points and the fitted line.

We need to find the values of α and β which minimize

$$g(\alpha, \beta) = \sum_{i=1}^n (y_i - \alpha - \beta x_i)^2$$

These values are determined by simultaneously solving the equations

$$\begin{aligned} \frac{\partial g}{\partial \alpha} &= \frac{\partial}{\partial \alpha} \sum_{i=1}^n (y_i - \alpha - \beta x_i)^2 = \sum_{i=1}^n 2(y_i - \alpha - \beta x_i)(-1) = 0 \\ \frac{\partial g}{\partial \beta} &= \frac{\partial}{\partial \beta} \sum_{i=1}^n (y_i - \alpha - \beta x_i)^2 = \sum_{i=1}^n 2(y_i - \alpha - \beta x_i)(-x_i) = 0 \end{aligned}$$

These equations can be written as

$$\bar{y} - \alpha - \beta \bar{x} = 0 \quad (1)$$

$$\sum_{i=1}^n (y_i - \alpha - \beta x_i)(x_i) = 0 \quad (2)$$

From equation (1), we obtain $\alpha = \bar{y} - \beta \bar{x}$ which we can substitute into equation (2) to obtain

$$\sum_{i=1}^n x_i [y_i - \bar{y} - \beta (x_i - \bar{x})] = 0$$

or

$$\beta = \frac{\sum_{i=1}^n x_i (y_i - \bar{y})}{\sum_{i=1}^n x_i (x_i - \bar{x})} = \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n (x_i - \bar{x}) (y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}$$

Therefore, the least squares estimates are

$$\alpha = \hat{\alpha} = \bar{y} - \hat{\beta} \bar{x} \quad \text{and} \quad \beta = \hat{\beta} = \frac{S_{xy}}{S_{xx}}$$

And, the equation of the fitted line is

$$y = \hat{\alpha} + \hat{\beta} x$$

0.1.4 STAT 231 Versus STAT 230 Final Grades

EXAMPLE 0.1.3: Fitted Line for STAT 230/231 Final Grades

For the STAT 230/231 data, we have the following

$$\bar{x} = \frac{2302}{30} = 76.7333 \quad \bar{y} = \frac{2167}{30} = 72.2333$$

$$S_{xx} = 5135.8667 \quad S_{xy} = 5106.8667 \quad S_{yy} = 7585.3667$$

$$\hat{\beta} = \frac{S_{xy}}{S_{xx}} = \frac{5106.8667}{5135.8667} = 0.9944$$

$$\hat{\alpha} = \bar{y} - \hat{\beta} \bar{x} = 72.2333 - (0.9944)(76.7333) = -4.0667$$

The fitted line is

$$y = -4.0667 + 0.9944x$$

Predicting STAT 231 Final Grade

Given your STAT 230 Final Grade what is the best estimate of your STAT 231 final grade based on these data?

If your final grade in STAT 230 was $x = 75$, then the least squares estimate of your STAT 231 final grade is

$$\hat{y} = -4.0667 + 0.9944(75) = 70.51$$

What can we say about the uncertainty in this estimate?

We need a statistical model in order to obtain an interval estimate of your mark, in particular we would like to construct a confidence interval.

We need a model which captures the fact that not everyone with a final grade of 75 in STAT 230 gets a final grade of 70.51 in STAT 231.

Determining a Model for STAT 230/231 Final Grades

Let's begin by considering the population of students who obtained a final grade of $x = 75$ in STAT 230.

Let $Y =$ STAT 231 final grade of a student drawn at random from this population.

What distribution might we assume for Y ?

Since frequency histograms for marks often exhibit a bell shape, we might assume $Y \sim G(\mu, \sigma)$ where μ represents the mean STAT 231 final grade for students in the study population who obtained a final grade of $x = 75$ in STAT 230.

We could then use this model along with the observed data to obtain interval estimates for the mean μ .

Linear Relationship Between STAT 230/231 Final Grades

In our sample of 30 students, we only observed one student with a final grade of 75 in STAT 230.

Does it make sense to do estimation with only one observation?

What to do? We do have 29 other observations.

Since the other 29 students had different STAT 230 final grades, they are observations drawn from populations which have different means (and possibly different variances).

From the scatterplot however, the relationship between STAT 230 and STAT 231 final grades look very linear.

Linear Relationship Between STAT 230/231 Final Grades: Note

Note we have assumed that the standard deviation σ does not depend on x_i .

We will look at ways of assessing whether this assumption is reasonable.

Model for STAT 230/231 Grades

It seems reasonable to assume a model in which the mean STAT 231 final grade for students in the study population who obtained a final grade of x in STAT 230 takes the form:

$$\mu(x) = \alpha + \beta x$$

For the data (x_i, y_i) , $i = 1, 2, \dots, n$, we assume the model

$$Y_i \sim G(\alpha + \beta x_i, \sigma)$$

for $i = 1, 2, \dots, n$ independently, where x_i is assumed to be a known constant.

This model is usually referred to as a **simple linear regression model**.

0.1.5 Simple Linear Regression Model

In the model, defined by

$$Y_i \sim G(\alpha + \beta x_i, \sigma)$$

for $i = 1, \dots, n$ independently where x_i is assumed to be a known constant, there are three unknown parameters: α , β , and σ .

For the STAT 230/231 data, the parameter $\mu(x) = \alpha + \beta x$ represents the mean STAT 231 final grade in the study population of students with a STAT 230 final grade equal to x .

Simple Linear Regression Model: Parameters α and β

What does the parameter β represent?

The parameter β represents the change in the mean $\mu(x) = \alpha + \beta x$ for a unit increase in x .

What does the parameter α represent?

The parameter α represents the mean in the study population of students with a STAT 230 final grade equal to 0. (Note: In this example, the parameter α is not of interest since students with a final grade of 0 cannot take STAT 231!)

0.1.6 Maximum Likelihood Estimates

Since our assumed model is

$$Y_i \sim G(\alpha + \beta x_i, \sigma), \quad \text{for } i = 1, \dots, n \text{ independently}$$

where x_i is assumed to be a known constant, the likelihood function for α and β (assuming for the moment that σ is known) is

$$\begin{aligned} \mathcal{L}(\alpha, \beta) &= \prod_{i=1}^n \exp \left[-\frac{1}{2\sigma^2} (y_i - \alpha - \beta x_i)^2 \right] \\ &= \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \alpha - \beta x_i)^2 \right] \end{aligned}$$

where we have ignored terms not involving α and β .

To obtain the maximum likelihood estimates of α and β , we would maximize

$$\mathcal{L}(\alpha, \beta) = \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \alpha - \beta x_i)^2 \right], \alpha \in \mathbb{R}, \beta \in \mathbb{R}$$

with respect to α and β , or equivalently minimize

$$\sum_{i=1}^n (y_i - \alpha - \beta x_i)^2$$

with respect to α and β .

This is a problem we've already solved!

Fitting a Straight Line: Maximum Likelihood Approach

Important Result

For the model

$$Y_i \sim G(\alpha + \beta x_i, \sigma), \text{ for } i = 1, 2, \dots, n \text{ independently}$$

where x_i is assumed to be a known constant, the maximum likelihood estimates of α and β (usually called the **regression parameters**) are given by

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} \quad \text{and} \quad \hat{\beta} = \frac{S_{xy}}{S_{xx}}$$

which are also the least square estimates of α and β .

So if the least squares estimates and the maximum likelihood estimates are the same, why do we actually assume a probability model? Why complicate things?

We need a probability model to model the variability in the data. Remember, not every student with a 75 in STAT 230 obtain the same STAT 231 final grade.

0.1.7 Distribution of the Maximum Likelihood Estimator of the Slope, Beta

In order to construct a confidence interval for your STAT 231 mark, we need to derive some distributional results.

Important Result

The first of these is

$$\tilde{\beta} \sim G\left(\beta, \frac{\sigma}{\sqrt{S_{xx}}}\right)$$

where

$$\tilde{\beta} = \frac{S_{xy}}{S_{xx}} = \frac{1}{S_{xx}} \sum_{i=1}^n (x_i - \bar{x}) Y_i$$

The distribution of $\tilde{\beta}$ is determined by the assumption

$$Y_i \sim G(\alpha + \beta x_i, \sigma), \quad \text{for } i = 1, 2, \dots, n \text{ independently}$$

where $x_i, i = 1, 2, \dots, n$, are assumed to be known constants.

Since

$$\tilde{\beta} = \frac{S_{xy}}{S_{xx}} = \frac{1}{S_{xx}} \sum_{i=1}^n (x_i - \bar{x}) Y_i$$

is a linear combination of independent normal random variables.

Thus, by a theorem learned in STAT 230, $\tilde{\beta}$ has a normal distribution.

We only need to find $\mathbb{E}[\tilde{\beta}]$ and $\mathbb{V}(\tilde{\beta})$.

$$\begin{aligned} \mathbb{E}[\tilde{\beta}] &= \sum_{i=1}^n \frac{(x_i - \bar{x})}{S_{xx}} \mathbb{E}[Y_i] \\ &= \sum_{i=1}^n \frac{(x_i - \bar{x})}{S_{xx}} (\alpha + \beta x_i) && \text{since } \mathbb{E}[Y_i] = \alpha + \beta x_i \\ &= \beta \sum_{i=1}^n \frac{(x_i - \bar{x})}{S_{xx}} x_i && \text{since } \sum_{i=1}^n \frac{(x_i - \bar{x})}{S_{xx}} \alpha = \frac{\alpha}{S_{xx}} \sum_{i=1}^n (x_i - \bar{x}) = 0 \\ &= \beta \frac{S_{xx}}{S_{xx}} \\ &= \beta \end{aligned}$$

Similarly,

$$\begin{aligned}
 \mathbb{V}(\tilde{\beta}) &= \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{(S_{xx})^2} \mathbb{V}(Y_i) \\
 &= \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{(S_{xx})^2} \sigma^2 && \text{since } \mathbb{V}(Y_i) = \sigma^2 \\
 &= \frac{\sigma^2}{(S_{xx})^2} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{\sigma^2}{(S_{xx})^2} S_{xx} \\
 &= \frac{\sigma^2}{S_{xx}}
 \end{aligned}$$

Thus,

$$\tilde{\beta} \sim G\left(\beta, \frac{\sigma}{\sqrt{S_{xx}}}\right)$$

0.1.8 Distribution of the Variance Estimator

Estimate of the Variance in Simple Linear Regression

Since σ is usually unknown, we estimate it using

$$s_e^2 = \frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{\alpha} - \hat{\beta}x_i)^2 = \frac{1}{n-2} (S_{yy} - \hat{\beta}S_{xy})$$

Note: $\sum_{i=1}^n (y_i - \hat{\alpha} - \hat{\beta}x_i)^2$ is usually called the **sum of the squared errors** and s_e^2 , the unbiased estimator, is called the **mean squared error**. s_e^2 is not the maximum likelihood estimate of σ^2 . The maximum likelihood estimate is similar, but with a denominator of n rather than $n-2$.

We use S_e^2 as the estimator of σ^2 since it can be shown that $\mathbb{E}[S_e^2] = \sigma^2$ where

$$S_e^2 = \frac{1}{n-2} \sum_{i=1}^n (Y_i - \tilde{\alpha} - \tilde{\beta}x_i)^2 \quad \tilde{\beta} = \frac{S_{xy}}{S_{xx}} \quad \tilde{\alpha} = \bar{Y} - \tilde{\beta}\bar{x}$$

Distribution of the Estimator of the Variance

It can be shown that

$$\frac{(n-2)S_e^2}{\sigma^2} = \frac{1}{\sigma^2} \sum_{i=1}^n (Y_i - \tilde{\alpha} - \tilde{\beta}x_i)^2 \sim \chi^2(n-2)$$

The proof is beyond the scope of this course, but it is usually proved in a third year linear regression course.

Note that there are $n-2$ degrees of freedom due to the two restrictions:

- $\sum_{i=1}^n (Y_i - \tilde{\alpha} - \tilde{\beta}x_i) = 0$
- $\sum_{i=1}^n (Y_i - \tilde{\alpha} - \tilde{\beta}x_i) x_i = 0$

Recall that these 2 equations 2 unknowns determined the estimators $\tilde{\alpha}$ and $\tilde{\beta}$.

0.1.9 Constructing the Confidence Interval for β

Since

$$\frac{\tilde{\beta} - \beta}{\sigma/\sqrt{S_{xx}}} \sim G(0, 1) \quad \text{and} \quad \frac{(n-2)S_e^2}{\sigma^2} \sim \chi^2(n-2)$$

independently, it follows that

$$\frac{\tilde{\beta} - \beta}{S_e/\sqrt{S_{xx}}} \sim t(n-2)$$

This t pivotal quantity can be used to construct confidence intervals and tests of hypotheses for β .

100p% Confidence Interval and p -Value for β

A 100p% confidence interval for β is given by

$$\hat{\beta} \pm a \frac{s_e}{\sqrt{S_{xx}}}$$

where $P(T \leq a) = \frac{(1+p)}{2}$ and $T \sim t(n-2)$.

For testing $H_0 : \beta = \beta_0$,

$$p\text{-value} = 2 \left[1 - P \left(T \leq \frac{|\hat{\beta} - \beta_0|}{s_e/\sqrt{S_{xx}}} \right) \right], \text{ where } T \sim t(n-2)$$

0.1.10 Hypothesis of No Relationship

Since $\mu(x) = \alpha + \beta x$, a test of $H_0 : \beta = 0$ is a test of the Hypothesis that the mean $\mu(x)$ does not depend on x .

This hypothesis is usually referred to as the hypothesis of no relationship between variates Y and x .

EXAMPLE 0.1.4: Estimates of β and σ for STAT 230/231 Final Grades

$$\begin{aligned} \bar{x} &= \frac{2302}{30} = 76.7333 & \bar{y} &= \frac{2167}{30} = 72.2333 \\ S_{xx} &= 5135.8667 & S_{xy} &= 5106.8667 & S_{yy} &= 7585.3667 \\ \hat{\beta} &= \frac{S_{xy}}{S_{xx}} = \frac{5106.8667}{5135.8667} = 0.9944 \\ s_e &= \sqrt{\frac{1}{n-2} (S_{yy} - \hat{\beta} S_{xy})} = \sqrt{\frac{1}{28} [7585.3667 - (0.9944)(5106.8667)]} = 9.4630 \end{aligned}$$

0.1.11 Interferences for the Slope

EXAMPLE 0.1.5: 95% Confidence Interval for β for STAT 230/231 Final Grades

Since

$$P(T \leq 2.0484) = \frac{1 + 0.95}{2} = 0.975 \quad \text{where } T \sim t(28)$$

a 95% confidence interval for β is

$$\hat{\beta} \pm 2.0484 \frac{s_e}{\sqrt{S_{xx}}} = 0.9944 \pm 2.0484 \frac{9.4630}{\sqrt{5135.8667}}$$

or

$$[0.7239, 1.2648]$$

EXAMPLE 0.1.6: Testing $H_0 : \beta = 0$ for STAT 230/231 Final Grades

Since the 95% confidence interval, $[0.7239, 1.2648]$, does not contain the value $\beta = 0$, the p -value for testing $H_0 : \beta = 0$ is smaller than 0.05.

This means there is evidence against the hypothesis of no relationship between STAT 231 final grades and STAT 230 final grades.

EXAMPLE 0.1.7: p -Value for STAT 230/231 Final Grades

The actual p -value for testing $H_0 : \beta = 0$ is

$$\begin{aligned} p\text{-value} &= 2 \left[1 - P \left(T \leq \frac{|\hat{\beta} - 0|}{s_e / \sqrt{S_{xx}}} \right) \right] \quad \text{where } T \sim t(28) \\ &= 2 \left[1 - P \left(T \leq \frac{0.9944}{9.4630 / \sqrt{5135.8667}} \right) \right] \\ &= 2[1 - P(T \leq 7.5304)] \\ &\approx 0 \end{aligned}$$

Therefore, there is very strong evidence against the hypothesis of no relationship between STAT 230 and STAT 231 final grades.

What does the hypothesis $H_0 : \beta = 1$ represent?

Recall, that the slope β represents the change in STAT 231 final grade for a unit change of 1 mark in the STAT 230 final grade in the study population.

EXAMPLE 0.1.8: Testing $H_0 : \beta = 1$ for STAT 230/231 Final Grades

Since the 95% confidence interval, $[0.7239, 1.2648]$, does contain the value $\beta = 1$, the p -value for testing $H_0 : \beta = 1$ is larger than 0.05 and there is no evidence against the hypothesis $H_0 : \beta = 1$.

The actual p -value for testing $H_0 : \beta = 1$ is

$$\begin{aligned} p\text{-value} &= 2 \left[1 - P \left(T \leq \frac{|\hat{\beta} - 1|}{s_e / \sqrt{S_{xx}}} \right) \right] \quad \text{where } T \sim t(28) \\ &= 2 \left[1 - P \left(T \leq \frac{|0.9944 - 1|}{9.4630 / \sqrt{5135.8667}} \right) \right] \\ &= 2[1 - P(T \leq 0.0428)] \end{aligned}$$

Since $P(T \leq 0.2558) = 0.6$,

$$p\text{-value} \geq 2(1 - 0.6) = 0.8$$

0.1.12 Interferences for the Mean Response at x

Suppose we wanted a confidence interval for the mean STAT 231 final grade for students who obtained a final grade of 75 in STAT 230; that is, we want a confidence for $\mu(75) = \alpha + \beta(75)$.

More generally we are often interested in a confidence interval for the mean response $\mu(x) = \alpha + \beta x$ for a specified value of x .

The maximum likelihood estimator of $\mu(x)$ is obtained by replacing the unknown values of α and β by their maximum likelihood estimators, which gives

$$\tilde{\mu}(x) = \tilde{\alpha} + \tilde{\beta}x = \bar{Y} + \tilde{\beta}(x - \bar{x})$$

since $\tilde{\alpha} = \bar{Y} - \tilde{\beta}\bar{x}$.

We will now show that

$$\begin{aligned}\tilde{\mu}(x) &= \tilde{\alpha} + \tilde{\beta}x \\ &= \bar{Y} + \tilde{\beta}(x - \bar{x}) \sim G\left(\mu(x), \sigma\sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}}\right)\end{aligned}$$

or

$$\tilde{\alpha} + \tilde{\beta}x \sim G\left(\alpha + \beta x, \sigma\sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}}\right)$$

We notice that the standard deviation is larger for values of x which are not close to the center or mean of the x data.

0.1.13 Distribution of the Estimator of the Mean Response at x

Recall our model assumption

$$Y_i \sim G(\alpha + \beta x_i, \sigma) \quad \text{for } i = 1, \dots, n \text{ independently}$$

where x_i is assumed to be a known constant, and the fact that

$$\tilde{\beta} = \frac{S_{xy}}{S_{xx}} = \sum_{i=1}^n \frac{(x_i - \bar{x})}{S_{xx}} Y_i$$

Then,

$$\tilde{\mu}(x) = \bar{Y} + \tilde{\beta}(x - \bar{x}) = \frac{1}{n} \sum_{i=1}^n Y_i + (x - \bar{x}) \sum_{i=1}^n \frac{(x_i - \bar{x})}{S_{xx}} Y_i = \sum_{i=1}^n a_i Y_i$$

where

$$a_i = \frac{1}{n} + (x - \bar{x}) \frac{(x_i - \bar{x})}{S_{xx}}$$

Since $\tilde{\mu}(x)$ is a linear combination of Gaussian random variables, it has a Gaussian distribution. We only need to find $\mathbb{E}[\tilde{\mu}(x)]$ and $\mathbb{V}(\tilde{\mu}(x))$. Facts:

$$\begin{aligned}\sum_{i=1}^n a_i &= 1 \\ \sum_{i=1}^n a_i x_i &= \sum_{i=1}^n \frac{1}{n} + \sum_{i=1}^n x_i (x - \bar{x}) \frac{(x_i - \bar{x})}{S_{xx}} = \bar{x} + (x - \bar{x}) \sum_{i=1}^n x_i \frac{(x_i - \bar{x})}{S_{xx}} = x\end{aligned}$$

Thus,

$$\begin{aligned}\mathbb{E}[\tilde{\mu}(x)] &= \sum_{i=1}^n a_i \mathbb{E}[Y_i] \\ &= \sum_{i=1}^n a_i (\alpha + \beta x_i) \\ &= \alpha \sum_{i=1}^n a_i + \beta \left(\sum_{i=1}^n a_i x_i \right) \\ &= \alpha + \beta x \\ &= \mu(x)\end{aligned}$$

$$\begin{aligned}
\mathbb{V}(\tilde{\mu}(x)) &= \sum_{i=1}^n a_i \mathbb{V}(Y_i) \\
&= \sigma^2 \sum_{i=1}^n a_i^2 \\
&= \sigma^2 \sum_{i=1}^n \left[\frac{1}{n^2} + \frac{2}{n} \frac{(x - \bar{x})(x_i - \bar{x})}{S_{xx}} + \frac{(x - \bar{x})^2 (x_i - \bar{x})^2}{(S_{xx})^2} \right] \\
&= \sigma^2 \left[\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}} \right]
\end{aligned}$$

Therefore, we have that

$$\mu(x) \sim G \left(\mu(x), \sigma \sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}} \right)$$

0.1.14 Interferences for the Mean

Since

$$\frac{\tilde{\mu}(x) - \mu(x)}{\sigma \sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}}} \sim G(0, 1) \quad \text{and} \quad \frac{(n-2)S_e^2}{\sigma^2} \sim \chi^2(n-2)$$

independently, it follows that

$$\frac{\tilde{\mu}(x) - \mu(x)}{S_e \sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}}} \sim t(n-2)$$

This pivotal quantity can be used to construct confidence intervals for $\mu(x)$ and to test hypotheses about $\mu(x)$.

Confidence Interval for the Mean Response at x

A 100p% confidence interval for $\mu(x) = \alpha + \beta x$ is given by

$$\hat{\mu}(x) \pm as_e \sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}} = \hat{\alpha} + \hat{\beta}x \pm as_e \sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}}$$

where $P(T \leq a) = \frac{1+p}{2}$ and $T \sim t(n-2)$.

EXAMPLE 0.1.9: 95% Confidence Interval for the Mean STAT 231 Final Grade

Since

$$P(T \leq 2.0484) = \frac{1+0.95}{2} = 0.975 \quad \text{where} \quad T \sim t(28)$$

a 95% confidence interval for the mean STAT 231 final grade for students who obtained a final grade of 75 in STAT 230 is

$$\begin{aligned}
\hat{\alpha} + \hat{\beta}x \pm as_e \sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}} &= -4.0667 + 0.9944(75) \pm 2.0484(9.4630) \sqrt{\frac{1}{30} + \frac{(75 - 76.7333)^2}{5135.8667}} \\
&= 70.51 \pm 3.5699
\end{aligned}$$

or

$$[66.9, 74.1]$$

Confidence Interval for the y -Intercept, α

Since $\mu(0) = \alpha + \beta(0) = \alpha$, a $100p\%$ confidence interval for α , is given by

$$\hat{\alpha} \pm as_e \sqrt{\frac{1}{n} + \frac{(\bar{x})^2}{S_{xx}}}$$

If $\bar{x}$ is large in magnitude (which means the average x_i is large), then the confidence interval for α will be very wide.

This would be disturbing if the value $x = 0$ is a value of interest, but it often not.

Confidence Interval for the Variance

The pivotal quantity

$$\frac{(n-2)S_e^2}{\sigma^2} \sim \chi^2(n-2)$$

can be used to construct intervals for σ^2 .

A $100p\%$ confidence interval for σ^2 is given by

$$\left[\frac{(n-2)s_e^2}{b}, \frac{(n-2)s_e^2}{a} \right]$$

where

$$P(U \leq a) = \frac{1-p}{2} = P(U > b), \quad \text{where } U \sim \chi^2(n-2)$$

A $100p\%$ confidence interval for σ is given by

$$\left[\sqrt{\frac{(n-2)s_e^2}{b}}, \sqrt{\frac{(n-2)s_e^2}{a}} \right]$$

0.1.15 Interference for an Individual Response Y at x

Suppose we wanted an interval for $Y = \text{STAT 231 final grade for one student who obtained a final grade of } x = 75 \text{ in STAT 230.}$

Now

$$\tilde{\mu}(x) = \tilde{\alpha} + \tilde{\beta}x \sim G \left(\alpha + \beta x, \sigma \sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}} \right)$$

and

$$Y \sim G(\alpha + \beta x, \sigma)$$

independently.

To obtain a confidence interval for Y , we first obtain the distribution of the random variable $Y - \tilde{\mu}(x)$ using its mean and variance.

Since

$$\mathbb{E}[Y - \tilde{\mu}(x)] = 0$$

and

$$\begin{aligned} \mathbb{V}(Y - \tilde{\mu}(x)) &= \mathbb{V}(Y) + \mathbb{V}(\tilde{\mu}(x)) \\ &= \sigma^2 + \sigma^2 \left[\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}} \right] \\ &= \sigma^2 \left[1 + \frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}} \right] \end{aligned}$$

we have

$$Y - \tilde{\mu}(x) \sim G\left(0, \sigma \left[1 + \frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}\right]^{1/2}\right)$$

Since

$$Y - \tilde{\mu}(x) \sim G\left(0, \sigma \left[1 + \frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}\right]^{1/2}\right)$$

and

$$\frac{(n-2)S_e^2}{\sigma^2} \sim \chi^2(n-2)$$

independently, we have

$$\frac{Y - \tilde{\mu}(x)}{S_e \sqrt{1 + \frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}}} \sim t(n-2)$$

This pivotal quantity can be used to construct a prediction interval for Y .

0.1.16 A $100p\%$ Prediction Interval for a Future Response Y

The corresponding interval

$$\hat{\mu}(x) \pm a s_e \sqrt{1 + \frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}} = \hat{\alpha} + \hat{\beta}x \pm a s_e \sqrt{1 + \frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}}$$

where a is the value from the t -table such that

$$P(T \leq a) = \frac{1+p}{2}, \quad \text{where } T \sim t(n-2)$$

We usually call such an interval a $100p\%$ prediction interval instead of a confidence interval, since Y is not a parameter but a random variable.

EXAMPLE 0.1.10: Prediction Interval Example

A 95% prediction interval for the STAT 231 final grade for a student who obtained a final grade of 75 in STAT 230 is

$$\begin{aligned} \hat{\alpha} + \hat{\beta}x \pm a s_e \sqrt{1 + \frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}} &= -4.0667 + 0.9944(75) \pm 2.0484(9.4630) \sqrt{1 + \frac{1}{30} + \frac{(75 - 76.7333)^2}{5135.8667}} \\ &= 70.51 \pm 19.7100 \end{aligned}$$

or

$$[50.8, 90.2]$$

0.1.17 Gaussian Response Models

The simple linear regression model we have just considered,

$$Y_i \sim G(\alpha + \beta x_i, \sigma), \quad \text{for } i = 1, 2, \dots, n \text{ independently}$$

where x_i is assumed to be a known constant, is a member of a larger family of models called **Gaussian response models**.

The general form of a Gaussian response model is

$$Y_i \sim G(\mu(x_i), \sigma), \quad \text{for } i = 1, 2, \dots, n \text{ independently}$$

where x_i is assumed to be a known constant, and where x_i can be a vector of explanatory variates also called **covariates**.

Note that the mean of Y_i depends on x_i , which may be a vector of variates (also called covariates in the linear regression model) or a scalar. However, the standard deviation σ does not depend on x_i .

The Gaussian response model

$$Y_i \sim G(\mu(x_i), \sigma), \quad \text{for } i = 1, 2, \dots, n \text{ independently}$$

can also be written in the form

$$Y_i = \mu(x_i) + R_i, \quad \text{where } R_i \sim G(0, \sigma), i = 1, 2, \dots, n \text{ independently}$$

Y_i is the sum of two components.

The first component, $\mu(x_i)$, is a deterministic component (not a random variable), and the second component R_i is a random component or random variable.

Linear Regression Models

In many examples,

$$\mu(x_i) = \beta_0 + \sum_{j=1}^k B_j x_{ij}$$

so the mean of Y_i is a linear function of $x_i = (x_{i1}, x_{i2}, \dots, x_{ik})$, the vector of covariates for unit i and the unknown parameters $\beta_0, \beta_1, \dots, \beta_k$.

These models are called **linear regression models**, and the B_j 's are called the **regression parameters**.

0.1.18 Model Checking

There are two main assumptions for Gaussian linear response models:

1. Y_i (given covariates x_i) is Gaussian with standard deviation σ which does not depend on covariates.
2. $\mathbb{E}[Y_i] = \mu(x_i)$ is a linear combination of observed covariates with unknown coefficients.

MODEL ASSUMPTIONS SHOULD ALWAYS BE CHECKED!!!

We will use graphical methods to do this.

Model Checking Method 1: Scatterplot with Fitted Line

In a simple linear regression, a scatter plot of the data with the fitted line superimposed shows us how well the model fits.

Model Checking Method 2: Residual Plots

Residual plots are very useful for model checking when there are two or more explanatory variates.

DEFINITION 0.1.11

For the simple linear regression model, let

$$\hat{\mu}_i = \hat{\alpha} + \hat{\beta}x_i$$

(often called the “fitted” response), and let

$$\hat{r}_i = y_i - \hat{\mu}_i, \quad i = 1, 2, \dots, n$$

The $\hat{r}_i$'s are **residuals** since $\hat{r}_i$ represents what is “left” after the model has been “fitted” to the data.

The $\hat{r}_i$'s can often be thought of as “observed” R_i 's in the model $Y_i = \mu_i + R_i$, where $R_i \sim G(0, \sigma)$, $i = 1, 2, \dots, n$ independently.

This isn't exactly correct since we are using $\hat{\mu}_i$ instead of μ_i , but if the model is correct, then the $\hat{r}_i$'s should behave roughly like a random sample from the $G(0, \sigma)$ distribution.

Recall $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$, which implies that $\bar{y} - \hat{\alpha} - \hat{\beta}\bar{x} = 0$ or

$$0 = \bar{y} - \hat{\alpha} - \hat{\beta}\bar{x} = \frac{1}{n} \sum_{i=1}^n (\bar{y}_i - \hat{\alpha} - \hat{\beta}\bar{x}_i) = \frac{1}{n} \sum_{i=1}^n \hat{r}_i$$

A plot of the points $(x_i, \hat{r}_i)$, $i = 1, 2, \dots, n$ should lie more or less within a horizontal band around the line $\hat{r}_i = 0$.

Standardized Residual Plots

DEFINITION 0.1.12

Define the **standardized residuals** as

$$\hat{r}_i^* = \frac{\hat{r}_i}{s_e} = \frac{y_i - \hat{\mu}_i}{s_e}, \quad i = 1, 2, \dots, n$$

What is the only difference between a plot of points $(x_i, \hat{r}_i)$, $i = 1, 2, \dots, n$ and a plot of points $(x_i, \hat{r}_i^*)$, $i = 1, 2, \dots, n$.

If the model is correct, then the $\hat{r}_i^*$ values will lie in the range of $(-3, 3)$. Why is this?

Model Checking Method 3: Plot of Residuals Versus Expected

Another type of residual plot can be used to check the assumption about the form of the mean. It consists of plotting the points $(\hat{\mu}_i, \hat{r}_i)$, $i = 1, 2, \dots, n$.

For the simple linear regression model, we are checking whether the assumed mean

$$\mathbb{E}[Y_i] = \mu(x_i) = \alpha + \beta x_i$$

is reasonable.

if the assumed mean is reasonable, we should see approximately a horizontal band around the line $\hat{r}_i = 0$.

A plot of $(\hat{\mu}_i, \hat{r}_i^*)$ looks like the plot of $(\hat{\mu}_i, \hat{r}_i)$. The only difference is that the vertical axis is rescaled.

Model Checking Method 4: Qqplot of Standardized Residuals

To check the normality assumption, we use a qqplot of the standardized residuals.

Since our assumed model is

$$\frac{Y_i - \mu_i}{\sigma} \sim G(0, 1)$$

the $\hat{r}_i^*$'s should represent a sample (not quite random since $\sum_{i=1}^n \hat{r}_i^* = 0$) from the $G(0, 1)$ distribution.

Therefore, a qqplot of the $\hat{r}_i^*$ terms should give approximately a straight line if the normality assumption holds.

Intepreting Residual Plots

If a plot of the points $(x_i, \hat{r}_i)$ or $(x_i, \hat{r}_i^*)$, $i = 1, 2, \dots, n$ shows a distinctive pattern, then this suggests the assumed form for $\mu(x_i)$ may be inappropriate.

If a plot of the points $(\hat{\mu}_i, \hat{r}_i)$, $i = 1, 2, \dots, n$ indicates that the variability in the $\hat{r}_i$'s is bigger for large values of $\hat{\mu}_i$ than for small values of $\hat{\mu}_i$ (or vice versa), then there is evidence to suggest that the assumption of constant variance $\mathbb{V}(Y_i) = \mathbb{V}(R_i) = \sigma^2$, $i = 1, 2, \dots, n$ does not hold.

Interpreting Residual Plots: Warning

Reading these plots takes practice. You should try not to read too much into plots especially if the plots are based on a small number of points.

The following plots exhibit patterns.

0.2 Comparing the Means of Two Populations

EXAMPLE 0.2.1: Hand Span Example

Suppose we wanted to answer the question: Are the hand spans of females enrolled in STAT 231 in Winter 2015 different on average from the hand spans of males enrolled in STAT 231 in Winter 2015? To do this we test the hypothesis that there is no difference in mean hand spans between males and females enrolled in STAT 231 in Winter 2015.

Let Y_{1i} = the hand span of the i th male, $i = 1, 2, \dots, 78$, and let Y_{2i} = the hand span of the i th female, $i = 1, 2, \dots, 64$.

Based on these observations, a Gaussian model seems reasonable for both the Y_{1i} 's and Y_{2i} 's.

Assume Y_{1i} , $i = 1, 2, \dots, 78$ is a random sample from a $G(\mu_1, \sigma)$ distribution, and independently Y_{2i} , $i = 1, 2, \dots, 64$ is a random sample from a $G(\mu_2, \sigma)$ distribution.

We call this a two sample Normal or Gaussian problem.

Note that we have assumed both Gaussian populations have the same standard deviation, σ .

There are three unknown parameters in the model: μ_1 , μ_2 , and σ .

The parameter μ_1 represents the mean hand span in centimeters for males enrolled in STAT 231 in Winter 2015 (the study population). Note that we are assuming there is no bias in the measurements due to the measurement system.

The parameter μ_2 represents the mean hand span in centimeters for females enrolled in STAT 231 in Winter 2015 (the study population).

The hypothesis of interest is $H_0 : \mu_1 = \mu_2$, or $H_0 : \mu_1 - \mu_2 = 0$.

0.2.1 Special Case of the Gaussian Response Model

By letting

$$Y_i = Y_{1i}, i = 1, 2, \dots, n_1 \quad \text{and} \quad Y_{n_1+i} = Y_{2i}, \quad i = 1, 2, \dots, n_2$$

$$\mathbb{E}[Y_i] = \mu_1, i = 1, 2, \dots, n_1 \quad \text{and} \quad \mathbb{E}[Y_{n_1+i}] = \mu_2, \quad i = 1, 2, \dots, n_2$$

and

$$\mathbb{V}(Y_i) = \sigma^2, \quad i = 1, 2, \dots, n_1 + n_2$$

we can see that this model is just a special case of the Gaussian response model.

$$Y_i \sim G(\mu(x_i), \sigma), \quad i = 1, 2, \dots, n \text{ independently}$$

where $\mu(x_i) = \mu_1$, $i = 1, 2, \dots, n_1$ and $\mu(x_i) = \mu_2$, $i = n_1 + 1, 2, \dots, n_1 + n_2$.

Likelihood Function for Random Samples from Two Gaussian Populations

The likelihood function for μ_1 , μ_2 , and σ is

$$L(\mu_1, \mu_2, \sigma) = \prod_{i=1}^{n_1} \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{1}{2} \left(\frac{y_{1i} - \mu_1}{\sigma} \right)^2 \right] \prod_{i=1}^{n_2} \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{1}{2} \left(\frac{y_{2i} - \mu_2}{\sigma} \right)^2 \right]$$

or more simply

$$L(\mu_1, \mu_2, \sigma) = \sigma^{-(n_1+n_2)} \exp \left[-\frac{1}{2} \sum_{i=1}^{n_1} \left(\frac{y_{1i} - \mu_1}{\sigma} \right)^2 \right] \exp \left[-\frac{1}{2} \sum_{i=1}^{n_2} \left(\frac{y_{2i} - \mu_2}{\sigma} \right)^2 \right]$$

$\mu_1 \in \mathbb{R}$, $\mu_2 \in \mathbb{R}$, and $\sigma > 0$.

Maximum Likelihood Estimators

The log likelihood function is

$$\ell(\mu_1, \mu_2, \sigma) = -(n_1 + n_2) \log \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^{n_1} (y_{1i} - \mu_1)^2 - \frac{1}{2\sigma^2} \sum_{i=1}^{n_2} (y_{2i} - \mu_2)^2$$

Maximization of this function gives the maximum likelihood estimators

$$\tilde{\mu}_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} Y_{1i} = \bar{Y}_1$$

$$\tilde{\mu}_2 = \frac{1}{n_2} \sum_{i=1}^{n_2} Y_{2i} = \bar{Y}_2$$

$$\tilde{\sigma}^2 = \frac{1}{n_1 + n_2} \left[\sum_{i=1}^{n_1} (Y_{1i} - \bar{Y}_1)^2 + \sum_{i=1}^{n_2} (Y_{2i} - \bar{Y}_2)^2 \right]$$

0.2.2 Pooled Estimator of Variance

Define

$$S_p^2 = \frac{1}{n_1 + n_2 - 2} \left[\sum_{i=1}^{n_1} (Y_{1i} - \bar{Y}_1)^2 + \sum_{i=1}^{n_2} (Y_{2i} - \bar{Y}_2)^2 \right]$$

Note that

$$S_p^2 = \left(\frac{n_1 - 1}{n_1 + n_2 - 2} \right) S_1^2 + \left(\frac{n_2 - 1}{n_1 + n_2 - 2} \right) S_2^2$$

where

$$S_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (Y_{1i} - \bar{Y}_1)^2$$

is the sample variance for the data from population 1, and

$$S_2^2 = \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (Y_{2i} - \bar{Y}_2)^2$$

is the sample variance for the data from population 2.

DEFINITION 0.2.2

$$\begin{aligned} S_p^2 &= \frac{1}{n_1 + n_2 - 2} \left[\sum_{i=1}^{n_1} (Y_{1i} - \bar{Y}_1)^2 + \sum_{i=1}^{n_2} (Y_{2i} - \bar{Y}_2)^2 \right] \\ &= \frac{(n_1 - 1) S_1^2 + (n_2 - 1) S_2^2}{n_1 + n_2 - 2} \end{aligned}$$

S_p^2 is called the **pooled estimator of variance**, since it is obtained by “pooling” the estimators S_1^2 and S_2^2 of σ^2 from the two samples.

Why does this estimator make sense?

The degrees of freedom are $n_1 + n_2 - 2$ because of the two restrictions

$$\sum_{i=1}^{n_1} (Y_{1i} - \bar{Y}_1) = 0 \quad \text{and} \quad \sum_{i=1}^{n_2} (Y_{2i} - \bar{Y}_2) = 0$$

It can also be shown that

$$\frac{(n_1 + n_2 - 2) S_p^2}{\sigma^2} = \frac{\sum_{i=1}^{n_1} (Y_{1i} - \bar{Y}_1)^2 + \sum_{i=1}^{n_2} (Y_{2i} - \bar{Y}_2)^2}{\sigma^2} \sim \chi^2 (n_1 + n_2 - 2)$$

0.2.3 Inferences for the Difference Between the Means

To make inferences about the mean difference $\mu_1 - \mu_2$, we note that $\tilde{\mu}_1 = \bar{Y}_1$ is a point estimator of μ_1 and $\tilde{\mu}_2 = \bar{Y}_2$ is a point estimator of μ_2 , so that $\tilde{\mu}_1 - \tilde{\mu}_2 = \bar{Y}_1 - \bar{Y}_2$ is a point estimator of $\mu_1 - \mu_2$.

Since

$$\bar{Y}_1 \sim N\left(\mu_1, \frac{\sigma^2}{n_1}\right) \quad \text{and} \quad \bar{Y}_2 \sim N\left(\mu_2, \frac{\sigma^2}{n_2}\right) \quad \text{independently}$$

we have

$$\tilde{\mu}_1 - \tilde{\mu}_2 = \bar{Y}_1 - \bar{Y}_2 \sim N\left(\mu_1 - \mu_2, \sigma^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)\right)$$

Pivotal Quantity for Confidence Interval for Mean Difference

Since

$$\frac{\bar{Y}_1 - \bar{Y}_2 - (\mu_1 - \mu_2)}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim G(0, 1)$$

and

$$\frac{(n_1 + n_2 - 2) S_p^2}{\sigma^2} \sim \chi^2 (n_1 + n_2 - 2)$$

independently, then

$$\frac{\bar{Y}_1 - \bar{Y}_2 - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t (n_1 + n_2 - 2)$$

and this is the pivotal quantity that we can use to make inferences about the mean difference $\mu_1 - \mu_2$ when σ is unknown.

Confidence Interval for Difference in Means

Since

$$\frac{\bar{Y}_1 - \bar{Y}_2 - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t (n_1 + n_2 - 2)$$

is a 100p% confidence interval for $\mu_1 - \mu_2$ is given by

$$\bar{y}_1 - \bar{y}_2 \pm as_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

where

$$P(T \leq a) = \frac{1+p}{2} \quad \text{and} \quad T \sim t (n_1 + n_2 - 2)$$

0.2.4 Test of Hypothesis for No Difference in Means

Since

$$\frac{\bar{Y}_1 - \bar{Y}_2 - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t(n_1 + n_2 - 2)$$

to test the hypothesis $H_0 : \mu_1 - \mu_2 = 0$, we use the test statistic

$$D = \frac{|\bar{Y}_1 - \bar{Y}_2 - 0|}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Let the observed value be

$$d = \frac{|\bar{y}_1 - \bar{y}_2 - 0|}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

then the p -value is

$$p\text{-value} = 2[1 - P(T \leq d)], \quad \text{where } T \sim t(n_1 + n_2 - 2)$$

EXAMPLE 0.2.3: Hand Span Example: Test of Hypothesis for No Difference in Means

For the males:

$$\hat{\mu}_1 = \bar{y}_1 = 21.50 \quad \text{and} \quad s_1^2 = 3.4309$$

For the females:

$$\hat{\mu}_2 = \bar{y}_2 = 19.37 \quad \text{and} \quad s_2^2 = 2.055$$

with

$$\hat{\mu}_1 - \hat{\mu}_2 = \bar{y}_1 - \bar{y}_2 = 21.50 - 19.37 = 2.14$$

and

$$s_p^2 = \frac{(77)(3.4309) + (63)(2.055)}{78 + 64 - 2} = 2.8117 \quad \text{and} \quad s_p = \sqrt{2.8117} = 1.6768$$

EXAMPLE 0.2.4: Hand Span Example: 95% Confidence Interval for $\mu_1 - \mu_2$

Using R, we obtain $P(T \leq 1.97705) = 0.975$, where $T \sim t(140)$. A 95% confidence interval for $\mu_1 - \mu_2$ is given by

$$\begin{aligned} \bar{y}_1 - \bar{y}_2 \pm as_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} &= 2.14 \pm (1.97705)(1.6768) \sqrt{\frac{1}{78} + \frac{1}{64}} \\ &= 2.14 \pm 0.5591 \\ &= [1.58, 2.70] \end{aligned}$$

Since $\mu_1 - \mu_2 = 0$ is not contained in this 95% confidence interval, we already know that the p -value for testing $H_0 : \mu_1 - \mu_2 = 0$ is less than 0.05.

Hand Span Example: p -Value

Since

$$d = \frac{|2.14 - 0|}{(1.67687) \sqrt{\frac{1}{78} + \frac{1}{64}}} = 7.56$$

the actual p -value is

$$\begin{aligned} p\text{-value} &= 2[1 - P(T \leq 7.56)], \quad \text{where } T \sim t(140) \\ &\approx 0 \end{aligned}$$

Therefore, there is very strong evidence to contradict the hypothesis $H_0 : \mu_1 - \mu_2 = 0$ based on the data.

The difference is statistically significant. Is the difference of practical significance?

0.2.5 Comparison of Two Means, Unequal Variances

Recall that the previous analysis depends on the assumptions

- $Y_{11}, Y_{12}, \dots, Y_{1n_1}$ is a random sample from $G(\mu_1, \sigma_1)$
- $Y_{21}, Y_{22}, \dots, Y_{2n_2}$ is a random sample from $G(\mu_2, \sigma_2)$
- the two samples are independent
- $\sigma_1 = \sigma_2$

What if $\sigma_1 = \sigma_2$ is not a reasonable assumption?

Note: $H_0 : \sigma_1 = \sigma_2$ could be tested using a likelihood ratio test.

Approximate Pivotal Quantity, Unequal Unknown Variances

If n_1 and n_2 are both large, then the approximate pivotal quantity is

$$\frac{\bar{Y}_1 - \bar{Y}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \sim G(0, 1) \text{ approximately}$$

This pivotal quantity can be used to construct confidence intervals and test hypotheses for the mean difference $\mu_1 - \mu_2$.

EXAMPLE 0.2.5

For example, an approximate 95% confidence interval for $\mu_1 - \mu_2$ would be given by

$$\bar{y}_1 - \bar{y}_2 \pm 1.96 \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

EXAMPLE 0.2.6: Hand Span Example: 95% Confidence Interval for $\mu_1 - \mu_2$

For the males we have the following:

$$n_1 = 78, \quad \hat{\mu}_1 = \bar{y}_1 = 21.50, \quad \text{and } s_1^2 = 3.4309$$

For the females we have the following:

$$n_2 = 64, \quad \hat{\mu}_2 = \bar{y}_2 = 19.37, \quad \text{and } s_2^2 = 2.055$$

An approximate 95% confidence interval for $\mu_1 - \mu_2$ is given by

$$\bar{y}_1 - \bar{y}_2 \pm 1.96 \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = 21.50 - 19.37 \pm 1.96 \sqrt{\frac{3.4309}{78} + \frac{2.055}{64}} = [1.60, 2.68]$$

as compared to $[1.58, 2.70]$.

0.3 Gaussian Response Models

0.3.1 Bean Experiment

In my Winter 2015 STAT 231 class, I conducted the following experiment.

Each student was given two small paper cups. One cup contained 30 black beans, and the other cup was empty.

Each student was also given a piece of paper on which two circles (the size of the paper cup bottom) were drawn.

Students were asked to place one paper cup on each circle so the cups were the same distance apart for each student, and then to use one hand to hold the cup containing the beans in place. Using the other hand, they were asked to move as many black beans as possible to the empty cup, one at a time, in a timed 15 -second interval. The students then performed the same task using the opposite hands.

Students were randomized with respect to whether they moved the beans with their dominant or nondominant hand first.

Bean Experiment Data

Difference in the Number of Beans Moved by the Dominant and Non-Dominant Hands

Difference	Frequency
-4	2
-3	1
-2	1
-1	4
0	13
1	15
2	14
3	5
4	2
Total	57

- Is there a difference in the mean number of beans moved in 15 seconds between the dominant and non-dominant hands?
- How do we analyze these data? What model should we assume? Is it a two sample problem?
- The assumptions for the two sample model are:

$Y_{11}, Y_{12}, \dots, Y_{1n_1}$ is a random sample from $G(\mu_1, \sigma)$

and independently

$Y_{21}, Y_{22}, \dots, Y_{2n_2}$ is a random sample from $G(\mu_2, \sigma)$

Bean Experiment: Correlation in the Data

Let Y_{1i} = number of beans moved using the dominant hand, and let Y_{2i} = number of beans moved using the non-dominant hand for the i th student.

Does it seem reasonable to assume the Y_{1i} 's are independent of the Y_{2i} 's?

We could expect the observations on the i th student (Y_{1i}, Y_{2i}) to be positively correlated. That is, we would expect $\text{Cov}(Y_{1i}, Y_{2i}) > 0$.

0.3.2 Paired Experiment

In fact, the observations from the bean experiment have been deliberately paired to eliminate some factors (e.g., finger size, agility, competitive spirit, etc.) which might otherwise affect conclusions about the parameter of interest, which is the mean difference $\mu_1 - \mu_2$

The bean experiment is an example of a **paired experiment**.

For a paired experiment (can you show this?)

$$\mathbb{V}(\bar{Y}_1 - \bar{Y}_2) = \frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{n} - \frac{2}{n}\text{Cov}(Y_{1i}, Y_{2i})$$

If $\text{Cov}(Y_{1i}, Y_{2i}) > 0$, then $\mathbb{V}(\bar{Y}_1 - \bar{Y}_2)$ is smaller than for an unpaired experiment.

Paired Experiment: Making Inferences about the Difference $\mu = \mu_1 - \mu_2$

To make inferences about $\mu = \mu_1 - \mu_2$, we analyze the within-pair differences

$$Y_i = Y_{1i} - Y_{2i}, \quad i = 1, 2, \dots, n$$

We assume

$$Y_i = Y_{1i} - Y_{2i} \sim G(\mu_1 - \mu_2, \sigma), \quad i = 1, 2, \dots, n$$

independently.

We can now use the one-sample analysis that we used previously for analysing a random sample from a $G(\mu, \sigma)$ distribution, with $\mu = \mu_1 - \mu_2$.

Bean Experiment: Testing the Null Hypothesis

For the bean data,

$$\bar{y} = 0.86 \quad \text{and} \quad s = \left[\frac{1}{56} \sum_{i=1}^{57} (y_i - \bar{y})^2 \right]^{1/2} = 1.66$$

To test $H_0 : \mu = 0$, we use the test statistic

$$D = \frac{|\bar{Y} - 0|}{S/\sqrt{n}}$$

with the observed value

$$d = \frac{|\bar{y} - 0|}{s/\sqrt{n}} = \frac{|0.86 - 0|}{1.66/\sqrt{57}} = 3.90$$

and p -value

$$p\text{-value} = 2[1 - P(T \leq 3.90)] \approx 0, \quad \text{where } T \sim t(56)$$

and there is strong evidence against $H_0 : \mu = 0$ based on the observed data.

Bean Experiment: Practical Versus Statistical Significance

The difference is statistically significant.

Is the difference of practical significance?

0.3.3 Examples of Experiments on Differences Between Means

Examples in which the parameter of interest is the mean difference $\mu_1 - \mu_2$

1. Test for the difference in execution time between two algorithms A and B with randomly generated data.
2. Test for a difference in the error rates or speeds of two algorithm designed for image resolution or character/speech recognition on many different scenarios/problems.
3. Test whether one numerical method for nonlinear optimization is faster than another on a large population of potential test functions.
4. Artificial intelligence: test whether one learning algorithm learns a task faster than another.

Experimental Design I

Generate or select n_1 random “problems” (for example, data sets to be sorted, functions to be minimized, images to be resolved), and compute the mean execution time $\bar{Y}_1$ of algorithm A.

Generate or select another n_2 ($n_1 = n_2$ possibly) random “problems”, and compute the mean execution time $\bar{Y}_2$ of algorithm B.

Estimate the difference as $\bar{Y}_1 - \bar{Y}_2$.

In this case, $\bar{Y}_1$ and $\bar{Y}_2$ are independent random variables, thus

$$\mathbb{E}[\bar{Y}_1 - \bar{Y}_2] = \mu_1 - \mu_2$$

and

$$\mathbb{V}(\bar{Y}_1 - \bar{Y}_2) = \mathbb{V}(\bar{Y}_1) + \mathbb{V}(\bar{Y}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

Experimental Design II

Generate or select n random “problems” (for example, data sets to be sorted, functions to be minimized, images to be resolved), and compute the mean execution time $\bar{Y}_1$ of algorithm A.

Compute the mean execution time $\bar{Y}_2$ of algorithm B on the same set of n problems.

Estimate the difference as $\bar{Y}_1 - \bar{Y}_2$.

In this case $\bar{Y}_1, \bar{Y}_2$ are dependent random variables, thus

$$\mathbb{E}[\bar{Y}_1 - \bar{Y}_2] = \mu_1 - \mu_2$$

and

$$\mathbb{V}(\bar{Y}_1 - \bar{Y}_2) = \frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{n} - \frac{2}{n} \text{Cov}(Y_{1i}, Y_{2i})$$

If $\text{Cov}(Y_{1i}, Y_{2i}) > 0$, then $\mathbb{V}(\bar{Y}_1 - \bar{Y}_2)$ is smaller for Design II.

0.3.4 Pairing as a Design Choice

We expect that the covariance between the execution times of algorithms on the same problem to be positively correlated (harder problems have longer execution times).

A sample of dependent pairs (Y_{1i}, Y_{2i}) is better than two independent random samples (one of Y_{1i} ’s and one of Y_{2i} ’s) for estimating $\mu_1 - \mu_2$, since the difference $\mu_1 - \mu_2$, can be estimated more accurately (shorter confidence intervals) if $\text{Cov}(Y_{1i}, Y_{2i}) > 0$.

Note: If $\text{Cov}(Y_{1i}, Y_{2i}) < 0$, then pairing is a bad idea since it increases the value of $\mathbb{V}(\bar{Y}_1 - \bar{Y}_2)$.

In a paired experiment, we do not assume that Y_{1i} and Y_{2i} are independent random variables. We do, however, assume the differences $Y_i = Y_{1i} - Y_{2i}, i = 1, 2, \dots, n$ are independent (all different problems).

Paired Versus Unpaired

When you see data from a comparative study (i.e., one whose objective is to compare two distributions, often through their means), you have to determine whether it involves paired data or not.

Of course, a sample of Y_{1i} ’s and Y_{2i} ’s cannot be from a paired study unless there are equal numbers of each, but if there are equal numbers, the study might be either “paired” or “unpaired”.

0.4 Multinomial Models and Goodness of Fit

0.4.1 Multinomial Models and Goodness of Fit

Is the distribution of colours uniform?

Smarties Data

Colour	Observed Number	Expected Number
Red	80	$610 \left(\frac{1}{8}\right) = 76.25$
Green	73	76.25
Yellow	77	76.25
Blue	107	76.25
Purple	61	76.25
Brown	73	76.25
Orange	72	76.25
Pink	77	76.25
Total	610	610

How do we conduct a formal test of the hypothesis that the distribution of different colours is uniform?

Model and Hypothesis

$$\begin{aligned}
 P(\text{observing the data } y_1, \dots, y_k; \theta_1, \dots, \theta_k) &= P(Y_1 = y_1, \dots, Y_k = y_k; \theta_1, \dots, \theta_k) \\
 &= \frac{n!}{y_1! \dots y_k!} \theta_1^{y_1} \dots \theta_k^{y_k}
 \end{aligned}$$

where $0 < \theta_j < 1$, $\sum_{j=1}^k \theta_j = 1$, $y_j = 0, 1, \dots$, and $\sum_{j=1}^k y_j = n$.

For our example, Y_j = number of Smarties of colour j , $j = 1, 2, \dots, 8$ and $k = 8$.

We want to test the hypothesis

$$H_0 : \theta_j = \frac{1}{k}, \quad \text{for all } j = 1, 2, \dots, k$$

0.4.2 Multinomial Likelihood Function

The multinomial likelihood function is

$$L(\theta_1, \theta_2, \dots, \theta_k) = \frac{n!}{y_1! y_2! \dots y_k!} \theta_1^{y_1} \theta_2^{y_2} \dots \theta_k^{y_k}$$

or more simply

$$L(\theta_1, \theta_2, \dots, \theta_k) = \theta_1^{y_1} \theta_2^{y_2} \dots \theta_k^{y_k} = \prod_{j=1}^k \theta_j^{y_j}$$

The maximum likelihood estimates are

$$\hat{\theta}_j = \frac{y_j}{n}, \quad j = 1, 2, \dots, k$$

and the maximum likelihood estimators are

$$\tilde{\theta}_j = \frac{Y_j}{n}, \quad j = 1, 2, \dots, k$$

Multinomial Likelihood Ratio Test Statistic

For testing $H_0 : \theta = \theta_0 = \left(\frac{1}{k}, \frac{1}{k}, \dots, \frac{1}{k}\right)$, we use

$$\Lambda(\theta_0) = -2 \log \left[\frac{L(\theta_0)}{\mathcal{L}(\tilde{\theta})} \right] = 2l(\tilde{\theta}) - 2l(\theta_0)$$

where

$$\tilde{\theta} = \left(\frac{Y_1}{n}, \frac{Y_2}{n}, \dots, \frac{Y_k}{n} \right)$$

Now

$$\begin{aligned}\frac{L(\theta_0)}{\mathcal{L}(\hat{\theta})} &= \left[\prod_{j=1}^k \left(\frac{1}{k} \right)^{Y_j} \right] \div \left[\prod_{j=1}^k \left(\frac{Y_j}{n} \right)^{Y_j} \right] \\ &= \prod_{j=1}^k \left(\frac{n/k}{Y_j} \right)^{Y_j} = \prod_{j=1}^k \left(\frac{E_j}{Y_j} \right)^{Y_j}, \quad \text{where } E_j = n \left(\frac{1}{k} \right) = \frac{n}{k}\end{aligned}$$

Note that Y_j is the observed number, and E_j is the expected number of observations in category j if H_0 is true.

Therefore, the likelihood ratio test statistic for testing $H_0 : \theta = \theta_0 = \left(\frac{1}{k}, \frac{1}{k}, \dots, \frac{1}{k} \right)$ is

$$\begin{aligned}\Lambda(\theta_0) &= -2 \log \left[\frac{L(\theta_0)}{\mathcal{L}(\hat{\theta})} \right] = -2 \log \left[\prod_{j=1}^k \left(\frac{E_j}{Y_j} \right)^{Y_j} \right] \\ &= 2 \sum_{j=1}^k Y_j \log \left(\frac{Y_j}{E_j} \right) \\ &= 2 \sum_{j=1}^k (\text{observed frequency}) \times \log \left(\frac{\text{observed frequency}}{\text{expected frequency}} \right)\end{aligned}$$

Why does this test statistic make sense? When does it take on large values?

When does it take on small values?

Behaviour of the Multinomial Test Statistic

If all of the observed frequencies (the Y_j 's) equal the expected frequencies under H_0 (the E_j 's), then $\log \left(\frac{Y_j}{E_j} \right) = 0$, and $\Lambda(\theta_0) = 0$; otherwise, $\Lambda(\theta_0) > 0$.

So large values of $\Lambda(\theta_0)$ give evidence against H_0 .

0.4.3 Distribution of the Multinomial Likelihood Ratio Test Statistic

Multinomial Likelihood Ratio Test Statistic

The likelihood ratio test statistic for testing $H_0 : \theta_j = \frac{1}{8}, j = 1, 2, \dots, 8$ is

$$\begin{aligned}\Lambda(\theta_0) &= -2 \log \left[\frac{L(\theta_0)}{\mathcal{L}(\hat{\theta})} \right] \\ &= 2 \sum_{j=1}^8 Y_j \log \left(\frac{Y_j}{E_j} \right) \\ &= 2 \sum_{j=1}^8 (\text{observed frequency}) \times \log \left(\frac{\text{observed frequency}}{\text{expected frequency}} \right)\end{aligned}$$

where Y_j is the observed number and E_j is the expected number of observations in category j if H_0 is true.

Distribution of the Multinomial Likelihood Ratio Test Statistic

Recall that if θ is a scalar, then $\Lambda(\theta_0)$ has approximately a $\chi^2(1)$ distribution for large n if $H_0 : \theta = \theta_0$ is true.

If θ is a vector, then $\Lambda(\theta_0)$ still has approximately a χ^2 distribution for large n if $H_0 : \theta = \theta_0$ is true, but the degrees of freedom change.

The degrees of freedom in the multiparameter case depend on both how many parameters are unknown in the original model, and how many parameters must be estimated under the null hypothesis.

The multinomial likelihood function $L(\theta_1, \theta_2, \dots, \theta_8)$ is a function of $8 - 1 = 7$ parameters.

Under the hypothesis $H_0 : \theta_j = \frac{1}{8}, j = 1, 2, \dots, 8$, there are no unspecified parameters and no parameters needed to be estimated.

Therefore, $\Lambda(\theta_0)$ has approximately a $\chi^2(7)$ distribution if $H_0 : \theta_j = \frac{1}{8}, j = 1, 2, \dots, 8$ is true.

Approximate p -Value for Smarties Data

The observed value of the likelihood ratio test statistic is

$$\begin{aligned}\lambda(\theta_0) &= 2 \sum_{j=1}^8 y_j \log\left(\frac{y_j}{e_j}\right), \quad \text{where } e_j = 610 \left(\frac{1}{8}\right) = 76.25 \\ &= 2 \left[80 \log\left(\frac{80}{76.25}\right) + \dots + 77 \log\left(\frac{77}{76.25}\right) \right] = 35.00\end{aligned}$$

The approximate p -value is

$$\begin{aligned}p\text{-value} &\approx P(W \geq 35), \quad \text{where } W \sim \chi^2(7) \\ &\approx 0\end{aligned}$$

and there is very strong evidence based on the observed data against the hypothesis of an equal number of each colour.

Note: The Chi-squared approximation is good when n is large and the expected frequencies under H_0 are all at least five.

0.4.4 Pearson's Chi-Squared Goodness of Fit Statistic

An alternative test statistic that was developed historically before the likelihood ratio test statistic is the Pearson Goodness of Fit Statistic:

$$D = \sum_{j=1}^k \frac{(Y_j - E_j)^2}{E_j}$$

Note that when $Y_j = E_j$ for all j , then $D = 0$. Otherwise, $D \geq 0$.

For large n , D and Λ are asymptotically equivalent and have the same asymptotic Chi-squared distribution.

The Pearson Goodness of Fit Test Statistic is more popular than the Likelihood Ratio Test.

0.5 Goodness of Fit Examples

0.5.1 Checking the Fit of the Model

In order to study a data set, we typically assume a model in which $(Y_1, Y_2, \dots, Y_n)$ is a random sample from a distribution which is a member of the family of models

$$f(y; \theta), \quad \text{for } \theta \in \Omega$$

It is important to check that the model adequately represents the variability in the data. This can be done by comparing the observed data $(y_1, y_2, \dots, y_n)$ with what we would expect to get using the model.

One way to do this is to compare the observed frequencies based on the data with the expected frequencies calculated using probabilities from the assumed model. If the model is suitable, then the observed and expected frequencies should be "close".

Discrete Data Example: Alpha-Particle Emissions

Recall a previous example regarding the data collected by the scientists, Rutherford and Geiger on the number of alpha-particles emitted during a fixed time interval.

We examined the fit of the Poisson model to these data by comparing the observed frequencies with the expected frequencies calculated assuming a Poisson model with mean equal to the sample mean.

Observed and Expected Frequencies Under Assumed Poisson Model

Number of Alpha-Particles Detected	Freq. f_j	Expected Freq. e_j
0	57	54.42
1	203	210.42
2	383	407.43
3	525	525.54
4	532	508.41
5	408	393.47
6	273	253.77
7	139	140.28
8	45	67.86
9	27	29.18
10	10	11.29
11+	6	5.77
Total	2608	2607.99

We decided based on the frequencies in this table that a Poisson model fits these data reasonably well.

The drawback of this method is we don't have a good way of deciding how "close" is good enough.

We will now see how to do a formal test of the hypothesis that the Poisson model fits these data.

Alpha-Particle Emissions: Test of Fit of Poisson Model

Our model is Multinomial $(n; \theta_0, \theta_1, \dots, \theta_{11})$, and $\sum_{j=0}^{11} \theta_j = 1$, which is a function of 11 parameters. H_0 : Data fit a Poisson model or, more specifically,

$$H_0 : \theta_j = \frac{\theta^j e^{-\theta}}{j!}, \quad j = 0, 1, \dots, 10$$

Under H_0 , there is one unknown parameter θ which must be estimated. Therefore, the degrees of freedom for the Chi-squared approximation for the likelihood ratio test statistic are $11 - 1 = 10$

Note that the expected frequencies are all at least five, so we can use the Chi-squared approximation to obtain the p -value. The observed value of the likelihood ratio statistic is

$$\begin{aligned} 2 \sum_{j=0}^{11} f_j \log \left(\frac{f_j}{e_j} \right) &= 2 \left[57 \log \left(\frac{57}{54.42} \right) + \dots + 6 \log \left(\frac{6}{5.77} \right) \right] \\ &= 14.01 \end{aligned}$$

with

$$p\text{-value} \approx P(W \geq 14.01) = 0.17, \text{ where } W \sim \chi^2(10)$$

and there is no evidence against the Poisson model.

The observed value of the Pearson Goodness of Fit Statistic is

$$\sum_{j=0}^{11} \frac{(f_j - e_j)^2}{e_j} = 12.98$$

with

$$p\text{-value} \approx P(W \geq 12.98) = 0.22, \quad \text{where } W \sim \chi^2(10)$$

and there is no evidence against the Poisson model.

What to Do if Expected Frequencies Are Not All at Least 5?

Suppose we have the following table of observed frequencies and expected frequencies calculated under the null hypothesis.

Category	Observed Number	Expected Number
1	53	50
2	21	25
3	11	12.5
4	8	6.25
5	4	3.125
6	3	3.125
Total	100	100

The last two categories have expected frequencies less than five, so it may not be appropriate to use the Chi-squared approximation.

Usually we collapse two or more adjacent categories with the smallest expected frequencies.

Category	Observed Number	Expected Number
1	53	50
2	21	25
3	11	12.5
4	8	6.25
≥ 5	7	6.25
Total	100	100

0.5.2 Two-Way Tables and Testing for Independence of Two Variates

Program/ Hometown	Canadian Hometown	Non-Canadian Hometown	Total
Computer Science	33	14	47
Non-Computer Science	39	46	85
Total	72	60	132

Is there a relationship between a student's program and their hometown?

Relative Risk

Previously, we summarized these data using relative risk as a numerical summary. Proportion of Computer Science students with Canadian hometown:

$$\frac{33}{47} = 0.7021$$

Proportion of non-Computer Science students with Canadian hometown:

$$\frac{39}{85} = 0.4588$$

Relative risk = $\frac{0.7021}{0.4588} = 1.53$ Students who have a Canadian hometown are about one and a half times more likely to be in a Computer Science program.

How could we test the hypothesis that a student's program is independent of their hometown?

Two-Way Tables and Testing for Independence of Two Variates

Suppose n individuals are classified according to two different variates which have two possible values.

	B	$\bar{B}$	Total
A	y_{11}	y_{12}	$r_1 = y_{11} + y_{12}$
$\bar{A}$	y_{21}	y_{22}	$n - r_1$
Total	$c_1 = y_{11} + y_{21}$	$n - c_1$	n

Let Y_{11} = number of $A \cap B$ outcomes, Y_{12} = number of $A \cap \bar{B}$ outcomes Y_{21} = number of $\bar{A} \cap B$ outcomes, and Y_{22} = number of $\bar{A} \cap \bar{B}$ outcomes. Then

$$(Y_{11}, Y_{12}, Y_{21}, Y_{22}) \sim \text{Multinomial}(n, \theta_{11}, \theta_{12}, \theta_{21}, \theta_{22})$$

The null hypothesis is that the variates A and B are independent, or

$$H_0 : P(A \cap B) = P(A)P(B)$$

Let $P(A) = \alpha$ and $P(B) = \beta$, then the null hypothesis may be written as

$$H_0 : \theta_{11} = \alpha\beta$$

Likelihood and Maximum Likelihood Estimators

Since $(Y_{11}, Y_{12}, Y_{21}, Y_{22}) \sim \text{Multinomial}(n, \theta_{11}, \theta_{12}, \theta_{21}, \theta_{22})$, the likelihood function is

$$L(\theta_{11}, \theta_{12}, \theta_{21}, \theta_{22}) = \frac{n!}{y_{11}!y_{12}!y_{21}!y_{22}!} \theta_{11}^{y_{11}} \cdot \theta_{12}^{y_{12}} \cdot \theta_{21}^{y_{21}} \cdot \theta_{22}^{y_{22}}$$

or more simply

$$L(\theta_{11}, \theta_{12}, \theta_{21}, \theta_{22}) = \theta_{11}^{y_{11}} \cdot \theta_{12}^{y_{12}} \cdot \theta_{21}^{y_{21}} \cdot \theta_{22}^{y_{22}}$$

The maximum likelihood estimates are

$$\hat{\theta}_{ij} = \frac{y_{ij}}{n}, \quad i = 1, 2 \text{ and } j = 1, 2$$

and the maximum likelihood estimators are

$$\tilde{\theta}_{ij} = \frac{Y_{ij}}{n}, \quad i = 1, 2 \text{ and } j = 1, 2$$

0.5.3 Parameter Estimation under the Null Hypothesis

If $H_0 : \theta_{11} = \alpha\beta$ is true, then the likelihood function under H_0 is

$$\begin{aligned} \mathcal{L}(\alpha, \beta) &= (\alpha\beta)^{y_{11}} [\alpha(1-\beta)]^{y_{12}} [(1-\alpha)\beta]^{y_{21}} [(1-\alpha)(1-\beta)]^{y_{22}} \\ &= \alpha^{y_{11}+y_{12}} (1-\alpha)^{y_{21}+y_{22}} \cdot \beta^{y_{11}+y_{21}} (1-\beta)^{y_{12}+y_{22}}, \quad 0 < \alpha < 1, \quad 0 < \beta < 1 \end{aligned}$$

which is maximized for

$$\hat{\alpha} = \frac{y_{11} + y_{12}}{n} = \frac{r_1}{n} \quad \text{and} \quad \hat{\beta} = \frac{y_{11} + y_{21}}{n} = \frac{c_1}{n}$$

The maximum likelihood estimators for α and β are

$$\tilde{\alpha} = \frac{Y_{11} + Y_{12}}{n} \quad \text{and} \quad \tilde{\beta} = \frac{Y_{11} + Y_{21}}{n}$$

Why do these estimates make sense?

Likelihood Ratio Test Statistic

$$= 2 \left[Y_{11} \log \left(\frac{Y_{11}}{E_{11}} \right) + Y_{12} \log \left(\frac{Y_{12}}{E_{12}} \right) + Y_{21} \log \left(\frac{Y_{21}}{E_{21}} \right) + Y_{22} \log \left(\frac{Y_{22}}{E_{22}} \right) \right]$$

which is of the form

$$2 \sum_{j=1}^k (\text{observed frequency}) \times \log \left(\frac{\text{observed frequency}}{\text{expected frequency}} \right)$$

Observed Likelihood Ratio Test Statistic

The observed value of the likelihood ratio test statistic is

$$\lambda = 2 \left[y_{11} \log \left(\frac{y_{11}}{e_{11}} \right) + y_{12} \log \left(\frac{y_{12}}{e_{12}} \right) + y_{21} \log \left(\frac{y_{21}}{e_{21}} \right) + y_{22} \log \left(\frac{y_{22}}{e_{22}} \right) \right]$$

Note that

$$e_{11} = n \left(\frac{r_1}{n} \right) \left(\frac{c_1}{n} \right) = \frac{r_1 c_1}{n}$$

and the other expected frequencies can be obtained by subtraction.

Two-Way Table: Observed and [Expected]

	B	$\bar{B}$	Total
A	y_{11} $[e_{11} = \frac{r_1 c_1}{n}]$	y_{12} $[e_{12} = r_1 - e_{11}]$	$r_1 = y_{11} + y_{12}$
$\bar{A}$	y_{21} $[e_{21} = c_1 - e_{11}]$	y_{22} $[e_{22} = r_2 - e_{21}]$	$n - r_1$
Total	$c_1 = y_{11} + y_{21}$	$n - c_1$	n

Degrees of Freedom for the Chi-Squared Approximation

What are the degrees of freedom for the Chi-squared approximation? How many parameters were there in the original model?

$$(Y_{11}, Y_{12}, Y_{21}, Y_{22}) \sim \text{Multinomial}(n; \theta_{11}, \theta_{12}, \theta_{21}, \theta_{22})$$

How many parameters under $H_0 : \theta_{11} = \alpha\beta$? Why do the degrees of freedom make sense?

Approximate p -Value

$$\begin{aligned} p\text{-value} &\approx P(W \geq \lambda), \quad \text{where } W \sim \chi^2(1) \\ &= 2[1 - P(Z \leq \sqrt{\lambda})], \quad \text{where } Z \sim G(0, 1) \end{aligned}$$

EXAMPLE 0.5.1: Program/Hometown Example: Two-Way Table

Program / Hometown	Canadian Hometown	Non-Canadian Hometown	Total
Computer Science	$e_{11} = \frac{47 \times 72}{132} = 25.64$	$e_{12} = 47 - 25.64 = 21.36$	47
Non-Computer Science	$e_{21} = 72 - 25.64 = 46.36$	$e_{22} = 60 - 21.36 = 38.64$	85
Total	72	60	132

Program/Hometown Example: p -Value

$$\lambda = 2 \left[33 \log \left(\frac{33}{25.64} \right) + 39 \log \left(\frac{39}{46.36} \right) + 14 \log \left(\frac{14}{21.36} \right) + 46 \log \left(\frac{46}{38.64} \right) \right] \\ = 7.38$$

$$p\text{-value} = 2[1 - P(Z \leq \sqrt{7.38})] = 2[1 - P(Z \leq 2.72)] \approx 0.0066$$

There is strong evidence based on these observed data against the hypothesis that a student's program is independent of their hometown.